

Bill of Materials — VBF Power-Tool Cell (V4)

BOM

#	Component	Material	Mass (g/cell)	Mass (kg/kWh)	Note / source
1	Positive electrode coating	NMC-type (Chen2020)	8.44	0.679	layer_kg_m2 0.08220 x area 0.1027 m2; binder/conductive fractions not in parameter set
2	Negative electrode coating	Graphite (Chen2020)	6.12	0.492	layer_kg_m2 0.05959 x area 0.1027 m2
3	Positive current collector	Al (2700 kg/m3)	4.44	0.357	16 um x 0.1027 m2 x 2700; density from parameter set
4	Negative current collector	Cu (8960 kg/m3)	11.04	0.889	12 um x 0.1027 m2 x 8960; density from parameter set
5	Separator	PP/PE-class (Chen2020)	0.26	0.021	layer_kg_m2 0.002525 x area 0.1027 m2
6	Electrolyte (fill estimate)	EC/EMC LiPF6-class, sigma 2.4 S/m	5.48	0.441	pore volume 4.57 mL x 1.2 g/cm3 (literature, annotated estimate); not in contract mass
	Subtotal (contract layer mass)		30.3	2.439	sum of rows 1-5; equals calc-energy mass_kg 0.0302999
	Subtotal (with electrolyte estimate)		35.78	2.88	rows 1-6
7	Enclosure / casing / tabs	Not modeled	N/A	N/A	no casing parameter in Chen2020 set; honest N/A
	Cell energy		12.42		r2_v4_energy.json:energy_wh (Wh)
	kg/kWh basis		mass(g)/1000 / (12.423 Wh/1000)		per component